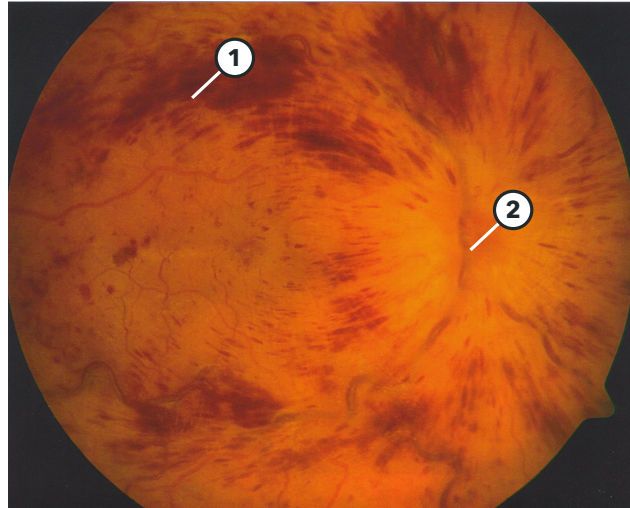


Retinal vascular disease

Clinical M3-M5 | Read the fundus pattern, identify the threat and choose the next step.

Read the pattern before naming the vessel

A diffuse haemorrhagic fundus



CRVO example. 1: haemorrhages in several retinal quadrants. 2: congested, swollen disc. Judge the whole distribution; an isolated haemorrhage is insufficient. Supplied M3 Anki, media 133.

Four patterns to separate

Pattern	Clinical implication
Diffuse haemorrhage + tortuous veins	CRVO; assess macular oedema and ischaemia
Haemorrhage confined to a venous sector	BRVO; map the affected drainage territory
Retinal whitening + red foveal contrast	CRAO; activate the acute stroke pathway
Microaneurysms/exudates ± surface new vessels	DR; grade retinopathy and assess the macula separately

A focused examination

Record onset and last-known-well time, acuity in each eye, RAPD, IOP and BP. Examine both fundi; check the iris/angle for new vessels when retinal ischaemia is suspected.

Systemic red flag

Systolic BP ≥ 180 or diastolic BP ≥ 120 mmHg with retinal haemorrhage or disc swelling requires same-day assessment for hypertensive organ injury.

References RCOphth: Retinal vein occlusion guideline | AAO: Retinal Artery Occlusions PPP (2025) | NICE NG136: Same-day hypertension assessment

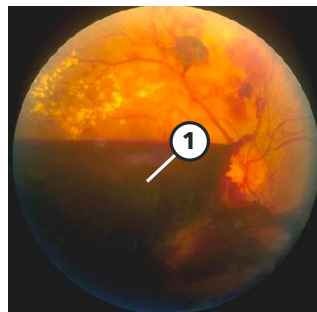
Diabetes: new vessels change the category

Proliferative disease at the disc



1: fine abnormal vessels on and around the optic disc establish PDR. 2: cotton-wool spot. Preserved acuity does not exclude dangerous proliferative disease. Supplied M3 Anki, media 48.

A second consequence of fragile new vessels



1: preretinal haemorrhage with a horizontal blood level. Fragile new vessels can cause sudden bleeding. Supplied M3 Anki, media 196.

Severe NPDR or PDR?

Category	Decisive finding
Severe NPDR	4-2-1: >20 haemorrhages in each of 4 quadrants; venous beading ≥ 2 ; or prominent IRMA ≥ 1 , without PDR
PDR	New vessels at disc/elsewhere, with or without preretinal/vitreous haemorrhage

Risk after proliferation

Surface new vessels can bleed; fibrovascular contraction can detach the retina. Refer promptly for retinal treatment planning.

Treat the macula and proliferation as two decisions

Macular leakage in diabetic eye disease



1: clustered hard exudates near the macula. 2: intraretinal haemorrhage. Exudates suggest prior leakage; OCT establishes current oedema and whether the centre is involved. Supplied M3 Anki, media 333.

What is threatening sight?

Problem	Next assessment / treatment
Centre-involving DME + reduced vision	Macular OCT; anti-VEGF is usually the main ocular treatment
Non-centre-involving DME	Retinal review; macular laser is appropriate in selected cases
PDR	PRP or anti-VEGF strategy, selected with the retina team
Macula-threatening traction / persistent dense bleed	Vitreoretinal assessment; vitrectomy may be needed

Attendance changes the choice

PRP offers a durable treatment effect and is especially important when repeated visits may be missed. Anti-VEGF can control PDR and DME but requires reliable follow-up; traction needs specialist caution.

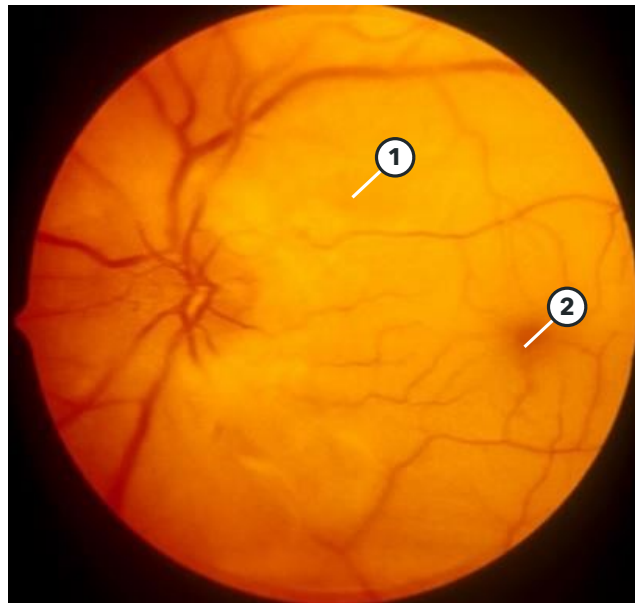
Systemic care still matters

Optimise glycaemia, BP and lipids, and maintain screening. DME can occur at any DR stage; good acuity alone does not grade retinopathy.

References ADA Standards of Care: Retinopathy (2026) | NICE NG242: Diabetic retinopathy | AAO: Diabetic Retinopathy PPP (2025)

CRAO: retinal ischaemia is a stroke emergency

A pale retina with a contrasting fovea



1: retinal whitening. 2: red foveal contrast, the cherry-red spot. The spot reflects background visibility and does not prove preserved central vision. Supplied M3 Anki, media 187.

Act immediately

- 1 Sudden severe painless monocular loss $\pm$ RAPD: record last-known-well time and examine promptly.
- 2 Suspected acute CRAO/BRAO: immediate emergency referral to a stroke-capable hospital; do not wait for outpatient angiography.
- 3 Stroke team evaluates acute treatment eligibility and carotid/cardiac sources; manage vascular risks and secondary prevention.

Look for giant cell arteritis

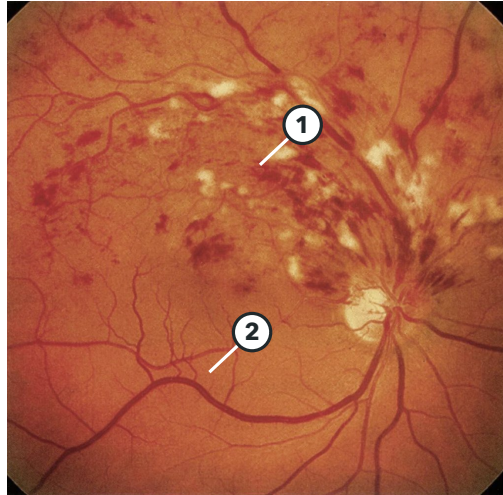
In people over 50, ask about new headache, jaw claudication and scalp tenderness. Strong suspicion needs urgent systemic corticosteroid treatment and GCA work-up without waiting for biopsy.

A resource correction

Routine ocular massage or anterior-chamber paracentesis has no established visual benefit and must not delay stroke assessment. Thrombolysis decisions belong to the acute stroke team and local protocol.

RVO: oedema and new vessels need different treatments

A venous sector rather than the whole retina



BRVO example. 1: haemorrhage/cotton-wool change in the affected sector. 2: relatively spared opposite sector. Compare this distribution with diffuse CRVO. Supplied M3 Anki, media 240.

Investigation follows the question

Question	Best tool / consequence
Is central blur due to fluid?	Macular OCT; anti-VEGF or selected steroid treatment for oedema
How much retina is non-perfused?	Fluorescein angiography or OCT angiography
Has anterior-segment NV developed?	Iris examination, IOP and gonioscopy; urgent retina/glaucoma care

Laser is not automatic

Do not routinely give prophylactic PRP to every CRVO eye. Ischaemic eyes need close surveillance; iris/angle new vessels call for prompt PRP with anti-VEGF as an adjunct. Selected high-risk eyes with unreliable follow-up may justify prophylactic treatment.

Check the person as well as the eye

Assess BP, diabetes, lipids and glaucoma risk. A painful red eye or rising IOP after RVO raises concern for neovascular glaucoma.